INCH-POUND
A-A-55618D
25 November 2003
SUPERSEDING
A-A-55618C
26 March 2002

COMMERCIAL ITEM DESCRIPTION

CASTER, SWIVEL, KINGPINLESS INDUSTRIAL, HEAVY DUTY WITH TOWING APPLICATION

This General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE: This Commercial Item Description (CID) covers industrial heavy duty kingpinless swivel casters for non-powered material transport and maintenance equipment with towing application for no more than five (5) miles per hour (mph).

2. SALIENT CHARACTERISTICS:

* 2.1 Casters furnished under this CID shall be flat mounted plate type kingpinless ball bearing swivel design. Fasteners used in the caster or brake assembly shall be of the locking type. Swivel casters shall conform to the dimensions and load rating as specified in Table I and figures below.

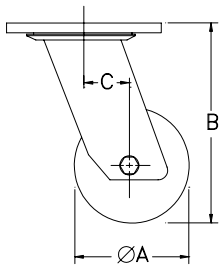


FIG. 1. SWIVEL CASTER

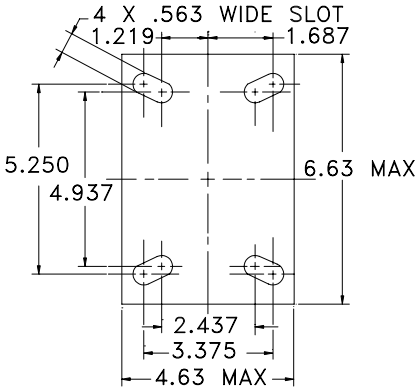


FIG. 2. MOUNTING PLATE

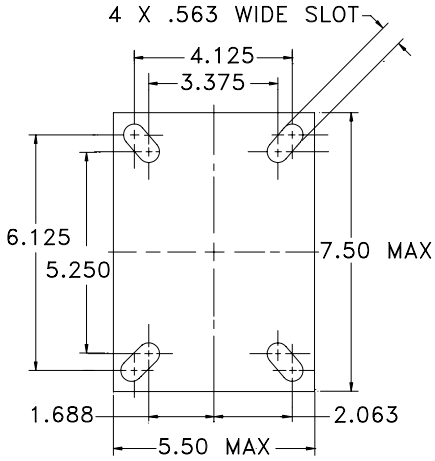


FIG. 3. MOUNTING PLATE

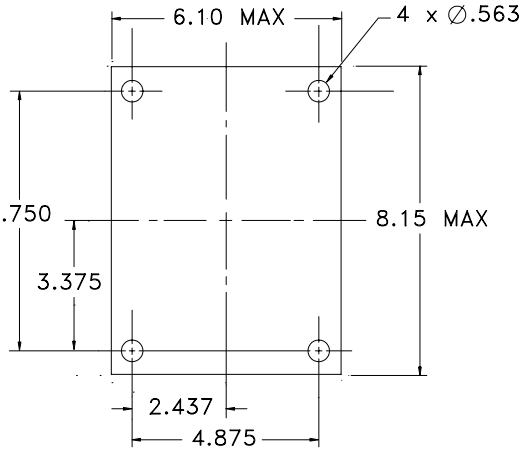


FIG. 4. MOUNTING PLATE

BENEFICIAL COMMENTS, RECOMMENDATIONS, ADDITIONS, DELETIONS, CLARIFICATIONS, ETC., AND DATA WHICH MAY IMPROVE THIS DOCUMENT SHOULD BE SENT TO: DEFENSE INDUSTRIAL SUPPLY CENTER, ATTN: DISC-EED, 700 ROBBINS AVENUE, PHILADELPHIA, PA 19111-5096.

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TABLE I. DASH NUMBERS AND DIMENSIONS.

DASH NO.	∅A	WHEEL WIDTH	WHEEL MATERIAL (SEE 2.2.6)	MIN. LOAD CAPACITY (LBS)	OVERALL HEIGHT B	MOUNTING BOLT DIAMETER	SWIVEL OFFSET C ± .25	MOUNTING PLATE
8	8.00	2.50	POLYURETHANE	1800	10.50	.500	2.50	SEE FIGURE 2
8H	8.00	2.50	POLYURETHANE	1800	11.00	.500	2.50	SEE FIGURE 3
12	12.00	3.00	POLYURETHANE	3400	14.75	.500	2.75	SEE FIGURE 4

2.1.1 Swivel caster. Swivel caster has a swivel horn and wheel assembly that will rotate freely through 360°.

2.2 Mounting plate and horn assembly. The mounting plate and horn assembly shall be made of steel plate, cast or forged steel. Legs shall be formed and welded both inside and out for additional strength. Mounting plate and legs shall have a minimum thickness of .375 inch.

2.2.1 Finish:

No code – Exposed metal surface of the caster assemblies shall be zinc plated per ASTM B633, Type II or Type III, SC2

Code G - Exposed metal surfaces of the caster assemblies shall be painted green, semi-gloss per MIL-PRF- 85285, Type II; color number 24052 per FED-STD-595.

2.2.2 Kingpinless raceway. Raceway shall be three (3) inches minimum diameter which measured from center of ball bearing to center of ball bearing at 180° apart.

2.2.3 Swivel assembly. The kingpinless swivel assembly shall be ball bearing assembly type with a smooth non- binding operation. The raceways shall be heat treated to a minimum hardness of 45 Rockwell C and shall be filled to capacity with .500 inch minimum diameter commercial quality steel ball bearings.

2.2.4 Wheel bearing. Wheel bearing shall be precision tapered roller or precision ball journal bearings having a bearing minimum life rating of 2500 hours at rated loads and rated speed and shall be manufactured in accordance with high-grade manufacturing practice. Wheel bearing shall be sealed with neoprene lip seal to retain grease and keep out dirt and contaminants.

2.2.5 Axle. The axle shall be .750 inch minimum diameter and shall be constructed of steel. The axle shall secure the wheel by means of a locknut or slotted nut and pin. Axle shall be hollow with lubrication fitting countersunk on the head to provide grease to wheel bearing. During operation, the axle shall not rotate loose or bind the wheel.

* 2.2.6 Wheel material. Wheel shall be solid metal core polyurethane material with shore A durometer hardness of 95 ± 5 points, and with crown type thread. Thread thickness shall be one (1) inch. Wheel material shall be able to withstand operating temperatures from –40 degrees F to 180 degrees F and ultraviolet light exposure without chipping, peeling, tearing, or flaking.

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2.2.7 Swivel lock. When specified, four position lever operated plunger type swivel lock shall be furnished. When engaged, the swivel lock shall prevent the caster from swiveling while rolling or shall prevent rolling while at right angles to the normal path of the equipment. When disengaged, the swivel lock shall permit full 360° free swiveling.

* 2.2.8 Brake. When specified, a manually operated with side or front mounted control lever parking brake shall be provided. The brake shall be tread contact type, using a combination of pressure and friction to prevent wheel rotation. The control lever shall be readily accessible for operation, permitting quick and easy brake application and release. The control lever shall not require a physical force in excess of 50 pounds to set the brake. When disengaged, the brake lever shall be capable of ensuring brake contact surface has a clearance of at least .062 inch off the wheel tread surface. When engaged, the brake shall be able to hold all caster wheels of the vehicle motionless, with the maximum gross load, whether headed up or down a 12 ±.5 degree incline. Brake shall be designed to withstand daily usage without repair or replacement for a minimum of five (5) years.

2.2.9 Lubrication fitting. The swivel assembly and wheel assembly shall have pressure lubrication fittings and shall be prelubricated with water repellent grease having a -32° C to +71° C (-10° F to +160° F) operating temperature range.

2.3 Casters as a minimum, shall be able to withstand the following tests:

2.3.1 Static load test. The caster shall be able to withstand four (4) times its rated capacity for one (1) minute minimum without damage.

2.3.2 Endurance test. The casters shall be loaded to rated capacity and shall be run continuously (see 4.3) at five (5) mph minimum on an endless belt or drum type testing machine with a smooth, uninterrupted running surface for a period of (40) hours. At the completion of the test, there shall be no indication of excess wear on the bearing races, looseness of bearings, scuffing of tires, looseness of tire tread, or permanently deformed members.

* 2.3.3 Obstacle test. The caster shall be loaded to rated capacity and shall be run continuously (see 4.3) at five (5) mph minimum on an endless belt or drum type testing machine for a minimum duration of two (2) hours. There shall be an obstacle (see 2.3.3.1) at every 6 feet of travel of the machine. At the completion of the test, there shall be no indication of excess wear to the bearing races, looseness of bearings, scuffing of tires, or looseness of tire tread, or permanently deformed members, or loss of any fastener on the caster or brake assembly. For caster equipped with brakes, the vibration and shock encountered during the test shall not cause the brake to engage or looseness of brake assembly.

2.3.3.1 Obstacles shall consists of a rectangular cross-sectioned steel bar .250 inch in thickness, so mounted that the longitudinal axis of the obstacles perpendicularly across the path of the casters.

* 2.3.4 Brake Test. For casters furnished with a brake, the brake must prevent wheel rotation from a horizontal pulling force equal to 120 percent (%) of the force required to initiate movement of the equipment under rated load with brake disengaged. The test shall be conducted on a level concrete floor by applying a horizontal pulling force on a rated load evenly distributed over four (4) casters, two of which will have the brakes engaged. The test should be repeated a minimum of four (4) times.

* 2.3.4.1 Brake Endurance Test. The brake assemblies shall be actuated a minimum of 4,000 cycles. Each cycle shall consists of engaging the brake, applying a 50-100 pounds horizontal force with the brake engaged, and then disengaging the brake. Upon completion of this test, the brake assemblies shall show no signs of excessive wear, loosening of fasteners, loss of parts, or permanently deformed members.

* 2.3.4.2 Brake Incline Test. The brake shall hold the vehicle motionless, with the maximum gross load, whether headed up or down a 11.5° incline plane. The test shall be conducted on a dry incline concrete floor with the rated load evenly distributed over four (4) casters, two of which will have the brakes engaged.

* 2.3.5 Impact Test. Caster shall be able to withstand impact delivered by a drop hammer at 2 times the rated load without damage.

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* 3. PRODUCT CONFORMANCE.

3.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and the quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

* 3.2 Certificates of Conformance. Manufacturer shall provide Certificates of Conformance for the caster working temperature and the ultraviolet light stabilizer for the wheel material.

4. NOTES.

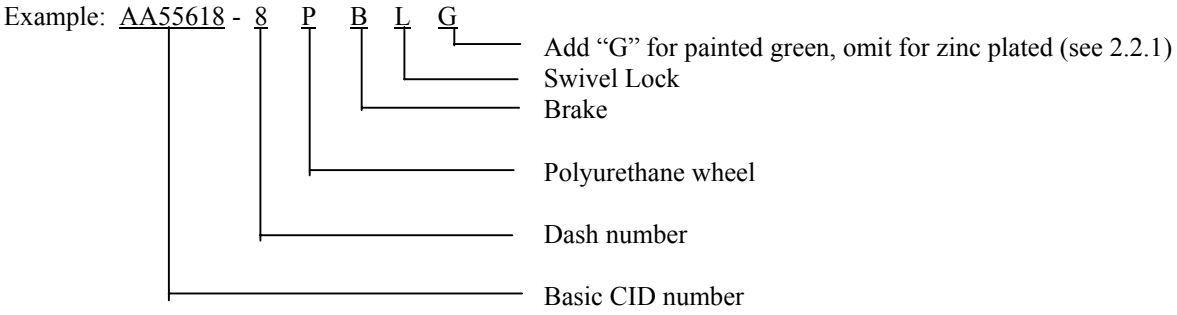
4.1 Unless otherwise specified, all dimensions are in inches.

4.2 Unless otherwise specified in the contract or purchase order, packaging shall be in accordance with ASTM D3951.

4.3 "Continuously" shall mean 45 minutes of rolling operation for each hour of work. The cycling of the stop and go periods shall be at the option of the manufacturer.

4.4 Unless otherwise specified, tolerances: .xx ± .12, .xxx ± .031.

4.5 The part identification number (PIN) shall consist of the basic commercial item description number followed by a dash number from Table I, a code letter for wheel material. Add "B" for Brake (optional) and/or "L" for swivel lock (optional) and/or "G " for painted finish (optional) after PIN number.



4.6 Cross-reference data. Rubber wheel caster PINs per rev B of this CID have been cancelled and are replaced by polyurethane wheel caster PINs.

Table II. Cross-reference table

Cancelled PIN	Replacement PIN
AA55618-8UBL	AA55618-8PBL
AA55618-8HUBL	AA55618-8HPBL
AA55618-12UBL	AA55618-12PBL

4.7 The margins of this specification are marked with asterisks to indicate where changes from the previous specification were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Navy – AS
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Reviewing Activity:
Air Force - 71

Civil Agency Coordination Activity:
GSA-FSS

Preparing Activity:
DLA-IS
(Project 5340-2721)

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